BRIDGES IN MATHEMATICS • GRADE 2 • UNIT 6

Geometry & Fractions: Partitioning Equal Shares

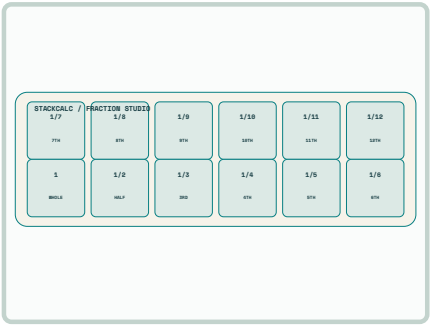
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.2.G.A.3

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Fraction Tray

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Place two 1/2 blocks (42mm) into the tray. Notice they match the 84mm whole block.

Sketch Model on Paper:
Draw a rectangle partitioned into 2 halves, 3 thirds, or 4 fourths.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE: FDISP Fraction Studio

1.	1/2	X: 1/2	Key 1st half (.1.2)
2.	ENTER	Y: 1/2 | X: 1/2	Stack lift
3.	1/2	Y: 1/2 | X: 1/2	Key 2nd half into X
4.	+	X: 1	Add: 1/2 + 1/2 = 1 w

EXPECTED LCD X: 1 (Whole)

1 Add three 1/3 blocks: 1/3 + 1/3 + 1/3.

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 Add four 1/4 blocks.

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 What remains if you remove 1/4 from 1 whole?

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 How many fourths equal 1/2?

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Unit 6

Geometry & Fractions: Partitioning Equal Shares • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Fraction Studio Proportional Towers

CAD Blueprint: fraction_tray

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/fraction_studio_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Teal 3x1 Bin & Color-Coded Fraction Towers (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Add three 1/3 blocks: 1/3 + 1/3 + 1/3.** Standard: CCSS.MATH.2.G.A.3	1	[.] [1] [.] [3] [ENTER] [.] [1] [.] [3
2	**Add four 1/4 blocks.** Standard: CCSS.MATH.2.G.A.3	1	[.] [1] [.] [4] [ENTER] [.] [1] [.] [4
3	**What remains if you remove 1/4 from 1 whole?** Standard: CCSS.MATH.2.G.A.3	3/4	[1] [ENTER] [.] [1] [.] [4] [-]
4	**How many fourths equal 1/2?** Standard: CCSS.MATH.2.G.A.3	2 fourths	[.] [1] [.] [2] [ENTER] [.] [1] [.] [4

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Equal shares must have identical volume/height, even if shape orientation differs.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.